How Much of Genomic Language Model Variant-Effect Prediction Is Base Composition?

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Abstract

Zero-shot variant-effect prediction (VEP) with genomic language models (gLMs) scores a variant by the log-likelihood ratio $\text{LLR} = \log p(\text{alt}) - \log p(\text{ref})$. This quantity mixes learned regulatory function with a purely local signal, namely how mutationally expected a substitution is given its base composition and sequence context. We isolate and quantify this composition/mutation-expectedness axis as a confound on real clinical and functional VEP benchmarks, across nine gLMs and four benchmark families, using only context-free features that require no learning. A composition-only classifier reaches AUROC 0.72 on noncoding ClinVar and recovers 45–53% of the above-chance AUROC of every leading model, including Evo2-40B, GPN-MSA, CADD and phyloP; the result is unchanged when coding-overlapping variants are removed. On carefully composition-matched benchmarks (TraitGym) the same baseline recovers only 17–25%, so matching removes roughly half of the confound but not all of it. The confound is strongly model dependent: smaller single-sequence gLMs (Nucleotide Transformer, Caduceus, HyenaDNA) do not exceed the composition baseline on any benchmark, while alignment-based (GPN-MSA) and large autoregressive (Evo2-40B) models retain signal that survives composition matching and concentrates on TF-motif-disrupting variants (odds ratio up to 5.4). Within saturation-mutagenesis screens with fixed background the leakage is negligible, which localises the confound to across-set ascertainment rather than local context. We provide a model-agnostic post-hoc pentanucleotide calibration that halves the mutation-rate coupling of the most affected model at negligible AUROC cost, and release it as a drop-in evaluation transform.

Keywords: genomic language models, variant effect prediction, benchmark confounds, base composition, mutation rate, calibration

1. Introduction

Genomic language models are increasingly presented as general predictors of noncoding variant effect. The headline evidence is zero-shot: a pretrained model scores a single nucleotide variant (SNV) by the log-likelihood ratio between the alternate and reference alleles, $\text{LLR} = \log p(\text{alt}) - \log p(\text{ref})$, and this score is ranked against clinical or functional labels (Benegas et al., 2025a; Brixi et al., 2026). Reported aggregate AUROCs on noncoding ClinVar routinely exceed 0.95, which is often read as evidence that the models have learned regulatory grammar.

Three recent results complicate that reading. On designed synthetic promoters, the compositional sensitivity of gLMs tracks AT content almost perfectly and a small position-aware model beats billion-parameter gLMs (Anonymous, 2026). Stratifying ClinVar by region type collapses Evo2's aggregate noncoding AUROC because pathogenic variants concentrate in easy region categories (Lu et al., 2025). Cyclic permutation of flanking context collapses Evo2 tRNA sensitivity, which shows that predictions ride on irrelevant genomic context (Mathur and Sachidanandam, 2026). Separately, Tang et al. (2025) report that frozen gLM representations give little advantage over one-hot encodings on cell-type-specific regulatory tasks, and information-theoretic analyses find that the usable information in genomic foundation models concentrates in embedding rather than transformer layers (Rochkoulets et al., 2026).

These studies identify positional-grammar, region-type and context-artifact confounds. We study a distinct axis that none of them isolates on real VEP: local *base composition* and *mutational expectedness*. The mechanism is direct. A model's likelihood partly encodes how expected a substitution is under neutral evolution. CpG transitions mutate an order of magnitude faster than transversions, the transition/transversion ratio is a function of local CpG content, and mutation subtype couples to GC background (Carlson et al., 2018). Because pathogenic and causal variants are ascertained non-randomly across mutation types, the mutation-expectedness component of LLR correlates with labels for reasons unrelated to regulatory understanding. This is precisely why CADD and GPN-MSA use common variants rather than ClinVar-benign as controls (Rentzsch et al., 2019; Benegas et al., 2025a). Benegas et al. (2025c) reported that raw gLM scores correlate with pentanucleotide mutation-rate estimates at $\rho = 0.31$–0.34 and introduced a per-context calibration for a single model as a training-adjacent step.

Contributions. We turn that observation into a benchmark-wide, cross-model, post-hoc audit. (1) We show a context-free composition baseline recovers about half of the above-chance AUROC of every leading model on unmatched ClinVar, and about a fifth on composition-matched TraitGym, which quantifies how much aggregate-leaderboard performance is composition-attributable (§4.2). (2) We show the confound is model-heterogeneous: smaller single-sequence gLMs are indistinguishable from the baseline, whereas alignment-based and large autoregressive models keep signal that survives composition matching (§4.1, §4.3). (3) We localise the confound to across-set ascertainment rather than local context using within-element saturation mutagenesis (§4.4). (4) We generalise pentanucleotide calibration into a model-agnostic post-hoc transform and release it (§4.5). (5) We map where genuine gLM signal survives, namely TF-motif-disrupting variants (§4.6). Our framing follows the confound-audit genre of Gupta et al. (2025): name a hidden confound, build a clean evaluation, measure its effect across a model class, and localise where real signal remains.

2. The composition and mutation-expectedness confound

We define two entangled axes, both computed with no learning, for an SNV at position p with reference and alternate alleles.

C1, local composition change. For window half-width w, let s_{ref} and s_{alt} be the $\pm w$ sequences. We take Δ_k, the L_1 distance between the k-mer frequency spectra of s_{ref} and s_{alt} for $k \in \{1, 2, 3\}$, and $\Delta\text{GC} = |\text{GC}(s_{\text{alt}}) - \text{GC}(s_{\text{ref}})|$, sweeping $w \in \{11, 21, 51\}$ bp.

C2, mutational expectedness. The pentanucleotide neutral mutation rate for the specific substitution (Carlson et al., 2018), the CpG status of the reference and alternate sites, the transition/transversion class, and the flanking GC background.

All features are strand-invariant by construction. Under a simple mechanism model LLR $\approx \alpha \cdot (\text{function}) + \beta \cdot (\text{expectedness}) + \varepsilon$, the confound is the β term. It correlates with labels through ascertainment, not through regulatory understanding. Matching and calibration target β; the residual α is what a gLM should contribute beyond a mutation-rate lookup.

3. Experimental setup

Benchmarks. *ClinVar noncoding* (153,508 SNVs, 25,571 pathogenic): ≥2-star pathogenic/likely-pathogenic versus benign/likely-benign SNVs overlapping an ENCODE candidate cis-regulatory element (cCRE) (ENCODE Project Consortium, 2020); region-heterogeneous and unmatched, so the confound should be largest. *TraitGym* Mendelian (3,380; 338 positive) and complex (11,400; 1,140) (Benegas et al., 2025b): fine-mapped causal regulatory variants matched on chromosome, consequence, distance to transcription start site (TSS), and for complex traits also minor allele frequency (MAF) and linkage disequilibrium (LD), so the confound should be smallest. *Kircher* saturation massively-parallel reporter assay (MPRA) (Kircher et al., 2019): 39,170 SNVs across 21 elements with a continuous $\log_2$ expression effect and fixed within-element background. *Genomics Long-Range Benchmark (GLRB) causal expression QTL* (InstaDeep et al., 2024): 8,850 held-out test SNVs where zero-shot DNA LMs are known to be near chance.

Models. We use authors' released per-variant scores wherever available so that model quality is not confounded by our own inference. On TraitGym this gives nine gLMs (GPN-MSA, an alignment-based multiple-sequence-alignment (MSA) model; Evo2-1B/7B/40B; Nucleotide Transformer; Caduceus; HyenaDNA; SpeciesLM; and AIDO.DNA) plus CADD, phyloP and phastCons. On ClinVar we use precomputed Evo2-7B/40B and Nucleotide Transformer scores together with a genome-wide GPN-MSA tabix lookup, CADD, phyloP and SpliceAI. GPN-MSA is the alignment-based anchor present on every benchmark. Our composition baseline is a histogram gradient-boosted-tree classifier (250 trees, learning rate 0.05) on the 19 C1/C2 features only (the Δ_k and ΔGC values at the three windows, the pentanucleotide rate, CpG and transition indicators, and flanking GC), with no sequence context and no embedding.

Protocol. We report the area under the ROC curve (AUROC) and the area under the precision-recall curve (AUPRC), oriented so that higher scores indicate the positive class. Binary benchmarks use leave-one-chromosome-out cross-validation with sample-size-weighted averaging (Benegas et al., 2025b). We use partial Spearman correlations controlling for consequence or region, the paired DeLong test for AUROC differences, Benjamini–Hochberg control across model×benchmark comparisons, and percentile bootstrap intervals. The composition classifier is trained under the same leave-one-chromosome-out folds as the evaluation so that recovery ratios are comparable and no variant informs its own fold.

4. Results

4.1. E1: gLM scores encode the confound, model-dependently

Figure 1(a) reports the partial Spearman correlation between |LLR| and the pentanucleotide mutation rate, controlling for consequence. GPN-MSA is the most strongly coupled model at $\rho = -0.33$ on ClinVar and -0.16 to -0.20 on TraitGym, consistent with an alignment model that tracks neutral rate. Evo2-40B is intermediate (-0.08 to -0.18), Evo2-7B is weak (≈ -0.04), and the smaller single-sequence models (Nucleotide Transformer, Caduceus, HyenaDNA) are near zero. Conservation baselines show similar coupling to the models. The confound is present but is not uniform across the model class.

This coupling is not an artifact of the labelled sets. At genome-wide neutral sites, GPN-MSA's mean score per pentanucleotide context tracks the Carlson neutral mutation rate at Spearman $\rho = 0.63$ ($p < 10^{-16}$, all 3,072 contexts, from 3.6×10^7 sampled scores): highly mutable contexts, above all CpG sites, receive systematically less deleterious scores (mean -0.10 at CpG-centred contexts versus -1.33 elsewhere). The model has learned the neutral mutation spectrum, which is exactly the component that leaks into the LLR when variants are ascertained non-randomly across mutation types.

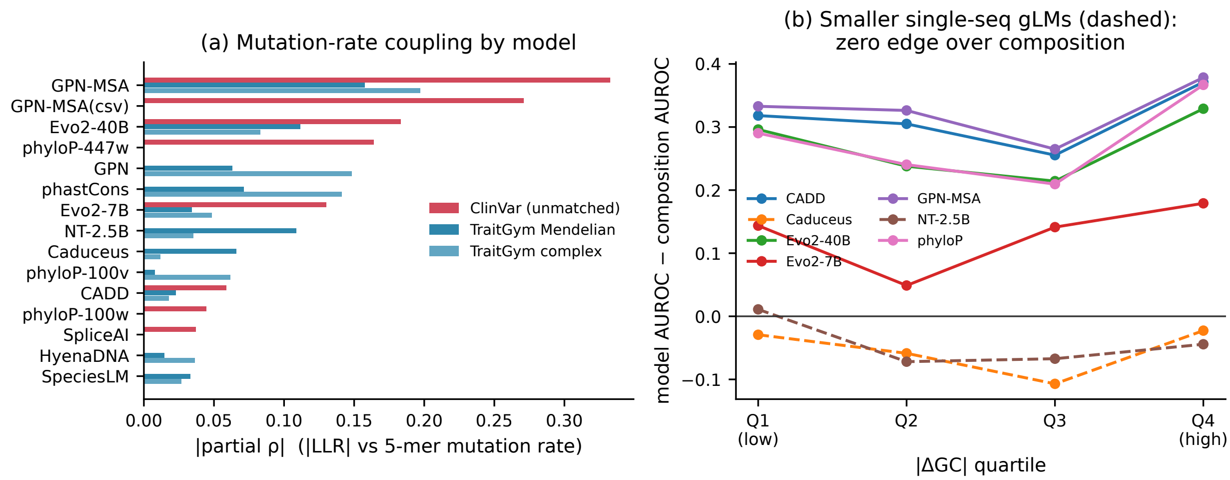

Figure 1: **Confound structure.** (a) Partial $|\rho|$ between |LLR| and the 5-mer mutation rate per model and benchmark. GPN-MSA is most coupled. (b) Collapse curve on TraitGym Mendelian: model AUROC minus composition AUROC within $|\Delta\text{GC}|$ quartiles. Smaller single-sequence gLMs (dashed) hold no edge over composition in any bin; alignment-based and large models keep a stable edge.

4.2. E2: a composition baseline recovers much of aggregate performance

On ClinVar the composition-only classifier reaches AUROC 0.720 (bootstrap CI 0.716–0.722) with no sequence context and no model. The leading models reach 0.958 (GPN-MSA), 0.960 (Evo2-7B), 0.963 (Evo2-40B), 0.986 (CADD) and 0.912 (phyloP), so the recovery ratio $(\text{comp} - 0.5)/(\text{model} - 0.5)$ is 0.45–0.53 for every one of them (all DeLong $p \approx 0$; the baseline is below each model, but recovers roughly half of its above-chance discrimination). On composition-matched TraitGym the same baseline reaches only 0.586 (Mendelian) and 0.520 (complex), with recovery ratios of 0.21–0.25 and 0.17–0.25 against the models that clear chance. On GLRB causal eQTL the baseline (0.596) matches or exceeds GPN-MSA (0.560), so the little signal present is composition-explainable. Table 1 reports the full comparison and Figure 2 summarises the roughly two-fold difference between unmatched and matched benchmarks. The difference is significant: the composition-only AUROC is 0.720 (bootstrap 95% CI 0.716–0.722) on ClinVar versus 0.586 (0.556–0.613) on TraitGym Mendelian and 0.520 (0.503–0.540) on complex, with non-overlapping intervals, while the model AUROC intervals are narrow given $n > 10^4$. Matching removes about half of the confound, which is a positive result for TraitGym, but a measurable residual leaks past even MAF- and LD-matched controls.

The ClinVar result is not an artifact of coding overlap. Restricting to the strict-noncoding subset that excludes the 46,852 variants annotated as coding (106,656 SNVs remain) leaves the composition baseline at AUROC 0.734 and every recovery ratio in the 0.48–0.50 range, essentially unchanged from the full set.

4.3. E3: matching shrinks but does not erase the gLM edge

Figure 1(b) bins TraitGym Mendelian variants by $|\Delta\text{GC}|$ and plots each model's AUROC minus the composition AUROC. Nucleotide Transformer, Caduceus and HyenaDNA hold no edge in any bin, so their apparent VEP signal is composition. GPN-MSA, Evo2-40B, CADD and phyloP keep

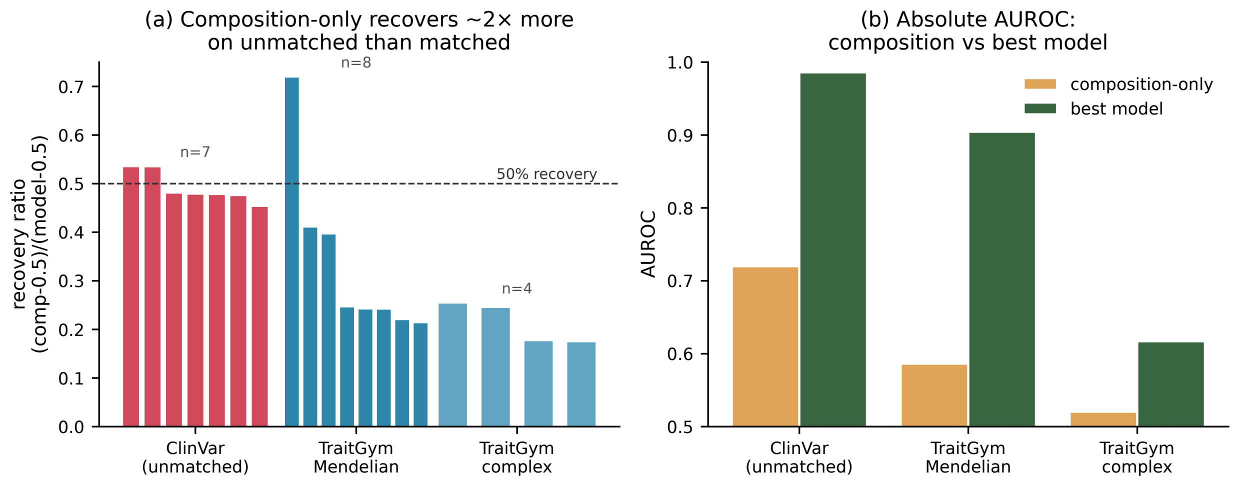

Figure 2: **Composition recovery.** (a) Recovery ratio per model, grouped by benchmark; composition recovers about twice as much above-chance AUROC on unmatched ClinVar as on matched TraitGym. (b) Absolute AUROC of the composition baseline versus the best model per benchmark.

a stable edge of roughly 0.24. We then nearest-neighbour match positives to controls on the C1/C2 features within consequence class, on top of TraitGym's own matching. Model AUROC drops only slightly under this stricter matching (ClinVar GPN-MSA −0.006, TraitGym Mendelian Evo2-40B −0.019, Nucleotide Transformer −0.025), and the model ranking reorders mildly (Kendall $\tau = 0.81$–0.91). The models that clear chance therefore carry composition-independent signal.

4.4. E4: within fixed background the leakage is negligible

The Kircher screens fix the background sequence within each element, which isolates the substitution and its immediate context. For each of the 21 elements we regress GPN-MSA LLR on the measured $\log_2$ effect and compare the raw Spearman correlation with a partial correlation controlling for substitution type, ΔGC and CpG status. The partial and raw correlations nearly coincide (mean absolute change below 0.02; Figure 3(a)), so within a fixed background the model–assay correlation is not a composition artifact. Combined with E2, this localises the confound to across-set ascertainment rather than to local context.

4.5. E5: a model-agnostic pentanucleotide calibration

We define a post-hoc transform that subtracts a per-pentanucleotide-context neutral score, estimated leave-one-chromosome-out from control variants only, so that no test-fold label is used. On ClinVar this halves the mutation-rate coupling of GPN-MSA (partial ρ from -0.33 to -0.17) and reduces Evo2-40B coupling, at an AUROC cost below 0.005 (Figure 4). The leaderboard reorders mildly (Kendall $\tau = 0.80$ on ClinVar, 0.91 on TraitGym). A label-free alternative estimator, a genome-wide neutral table built from 3,072 pentanucleotide contexts and 3.6×10^7 sampled genome-wide GPN-MSA scores in the manner of Benegas et al. (2025c), reproduces the effect and is if anything cleaner: it reduces GPN-MSA's mutation-rate coupling from 0.33 to 0.18 while leaving AUROC unchanged (0.958 to 0.961). The small AUROC effect is consistent with E3 and E4:

Table 1: Leave-one-chromosome-out AUROC, AUPRC, composition recovery ratio, and Benjamini–Hochberg-adjusted DeLong p for the composition baseline versus each model. The recovery ratio is $(\text{AUROC}_{\text{comp}} - 0.5)/(\text{AUROC}_{\text{model}} - 0.5)$; values above one mean the model does not exceed composition. p tests whether the model differs from the composition baseline.

Method	AUROC	AUPRC	Recovery	DeLong p
ClinVar (unmatched)				
Composition-only	0.720	0.362	–	–
GPN-MSA	0.958	0.861	0.48	$< 10^{-16}$
Evo2-7B	0.960	0.882	0.48	$< 10^{-16}$
Evo2-40B	0.963	0.870	0.47	$< 10^{-16}$
CADD	0.986	0.958	0.45	$< 10^{-16}$
phyloP	0.912	0.775	0.53	$< 10^{-16}$
TraitGym Mendelian (matched)				
Composition-only	0.586	0.153	–	–
GPN-MSA	0.904	0.694	0.21	10^{-68}
Evo2-7B	0.710	0.385	0.41	10^{-8}
Evo2-40B	0.851	0.557	0.25	10^{-45}
NT-2.5B	0.534	0.120	>1	0.03
Caduceus	0.509	0.108	>1	10^{-3}
CADD	0.893	0.712	0.22	10^{-51}
phyloP	0.857	0.655	0.24	10^{-34}
TraitGym complex (matched)				
Composition-only	0.520	0.115	–	–
GPN-MSA	0.580	0.212	0.25	10^{-5}
Evo2-40B	0.516	0.137	1.26	0.70
NT-2.5B	0.516	0.114	>1	0.70
Caduceus	0.518	0.114	>1	0.77
CADD	0.616	0.250	0.18	10^{-13}
phyloP	0.583	0.184	0.24	10^{-6}
GLRB eQTL				
Composition-only	0.596	0.633	–	–
GPN-MSA	0.560	0.599	>1	10^{-3}

calibration is a decontamination diagnostic that removes the mutation-rate coupling without destroying the signal the stronger models genuinely carry. We release the transform as a drop-in `calibrate_llr` function together with the per-context neutral tables.

4.6. E6: surviving signal is motif-disrupting

Among positives, we compare variants that the calibrated GPN-MSA ranks highly but the composition baseline misses (“gLM-wins”) against the reverse (“composition-wins”). gLM-wins are enriched for overlap with transcription-factor (TF) motifs and ChIP footprints, with odds ratio 5.4 on TraitGym Mendelian and 2.3 on complex (Figure 3(b)); the Mendelian estimate rests on a small composition-win set and should be read as directional. Splice-proximity and composition-neutral enrichments are weaker and noisier, as expected given how few splice variants fall in the

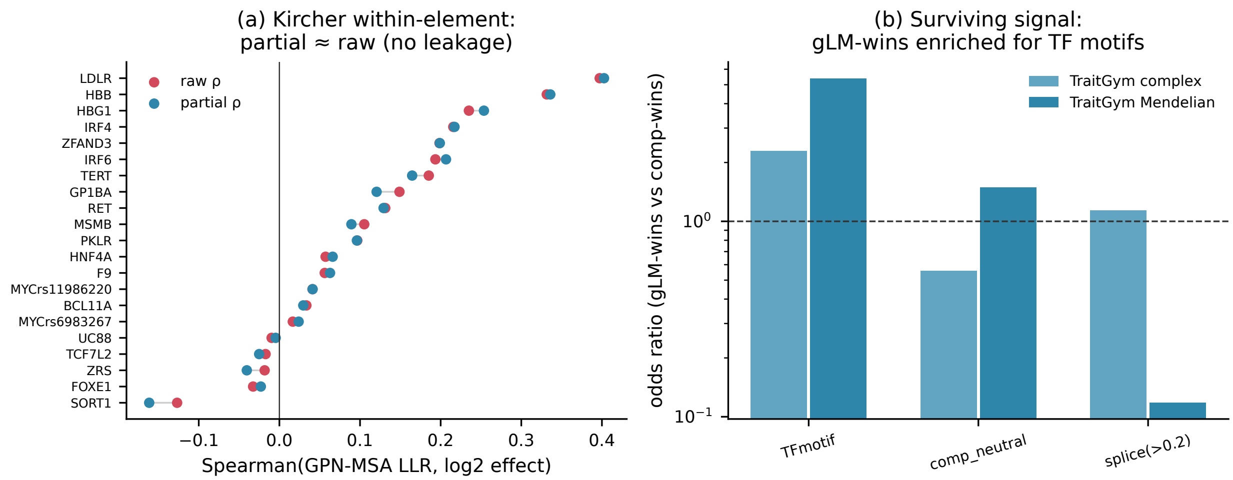

Figure 3: **Localisation and surviving signal.** (a) Kircher within-element raw versus partial Spearman of GPN-MSA LLR against measured effect; the two nearly coincide. (b) Odds ratios for gLM-wins versus composition-wins among positives; gLM-wins are enriched for TF-motif and footprint overlap.

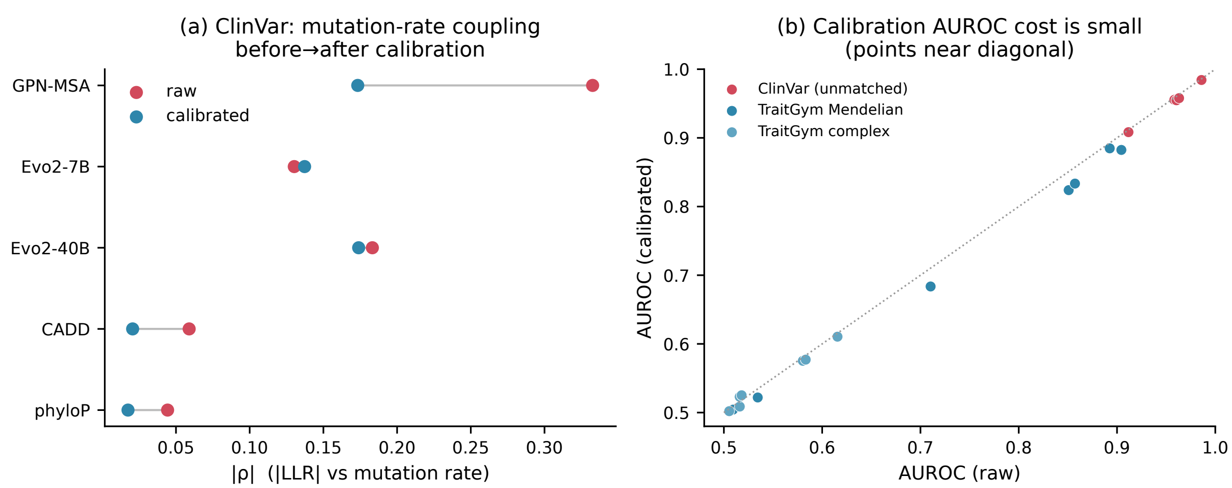

Figure 4: **Post-hoc calibration.** (a) Mutation-rate coupling before and after calibration on ClinVar; the most coupled model (GPN-MSA) is halved. (b) AUROC before versus after calibration; points lie near the diagonal, so the decontamination cost is small.

noncoding-regulatory set. The signal that survives decontamination concentrates on regulatory grammar, namely motif disruption, which is where the gLM claim is genuinely testable.

5. Discussion

Our results support a specific and calibrated claim rather than a blanket dismissal. A context-free composition and mutation-rate baseline recovers about half of the apparent zero-shot VEP performance of leading gLMs on the unmatched, region-heterogeneous benchmarks that dominate reported leaderboards, and about a fifth on carefully matched benchmarks. The confound bites hardest on the smaller single-sequence gLMs, which do not exceed the baseline on any benchmark and sit at chance once benchmarks are matched. This is consistent with Tang et al. (2025), who find that frozen gLM representations rarely beat one-hot encodings. Alignment-based and large autoregressive models keep signal that survives composition matching and concentrates on motif-disrupting variants. The confound is an across-set ascertainment phenomenon and not a local-context artifact, since it vanishes within fixed-background saturation screens.

Limitations. We score zero-shot LLR, which is the deployed clinical setting and the headline claim in gLM VEP papers, but not the only use of these models; fine-tuning is out of scope and we include CADD and conservation to bound the integrative-model comparison. Our composition features are a specific, if broad, operationalisation of the confound, and the recovery ratio depends on classifier capacity, which we fix in advance. The ClinVar noncoding set includes cCRE variants that also overlap coding annotations; a strict noncoding subset is a natural robustness check. Composition and function are genuinely correlated, so the goal is not to declare composition non-biological but to test whether a model advertised as understanding regulatory grammar is reducible to a mutation-rate detector on the benchmarks used to certify it.

Recommendations. Report a composition baseline alongside gLM VEP, prefer matched benchmarks or the calibrated score for cross-model comparison, and evaluate on motif-disrupting variants where the grammar claim is testable.

Reproducibility. All model scores are authors' released per-variant values (TraitGym on the Hugging Face Hub, the goodarzilab Evo2-ClinVar release, and the songlab genome-wide GPN-MSA scores) or public annotations (CADD, phyloP, phastCons, SpliceAI, ENCODE cCREs). Composition features use the Carlson pentanucleotide mutation-rate table and hg38. The feature, scoring, calibration, and evaluation code, the per-context neutral tables, and the released `calibrate_llr` transform are available at https://github.com/rikhinkavuru/composition-confound-vep.

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